

GAWTAM CHITHRA RAMESH

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Education

KTH Royal Institute of Technology

Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0)

Aug 2024 – Jun 2026

Stockholm, Sweden

Indian Institute of Technology Madras

Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10)

Aug 2019 – Jul 2024

Chennai, India

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, C, ROS2, MoveIt, Gazebo, Embedded Systems, PREEMPT-RT, SocketCAN, PyTorch, Unity3D, Fusion360, Docker, Git

Professional Experience

Master's Thesis: Generative Flow Matching for Motion Planning

Jan 2026 – Jun 2026

ABB Robotics

Västerås, Sweden

- Designed a generative flow-matching planner guided at inference by Signal Temporal Logic robustness gradients.
- Steered the pretrained flow model toward new temporal-logic specifications at inference without retraining.
- Benchmarked on long-horizon reach-avoid navigation against state-of-the-art specification-guided motion planners.

Device Research Intern: 3D Scene Graphs

Jul 2025 – Dec 2025

Ericsson Research

Lund, Sweden

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph construction framework.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.
- Benchmarked state-of-the-art inpainting models to balance reconstruction fidelity against real-time edge latency.

Graduate Robotics Intern: Collaborative Robots

Dec 2022 – Jul 2023

Systemantics India

Bangalore, India

- Built a complete real-time system controller from scratch for a 6-DoF cobot portfolio on a Linux PREEMPT-RT kernel.
- Handled inter-process communication over SocketCAN and FDCAN buses using semaphores and signal interrupts.
- Integrated a ROS2 stack with MoveIt2 motion planning and Gazebo simulation, authoring URDF and SRDF descriptions.

Research Experience

Autonomous Ultrasound via Single-Camera 3D Reconstruction

Jan 2024 – May 2024

Nirav Patel, Engineering Design, IIT Madras

Chennai, India

- Reconstructed a dense 3D torso point cloud from 20 RGB poses via structure-from-motion and multi-view stereo.
- Segmented the reconstructed point cloud to estimate probe approach angles from local surface tangent vectors.
- Closed the perception-to-planning loop with ROS and MoveIt motion planning validated in Gazebo physics simulation.

Target-Driven UAV Motion Planning and Obstacle Avoidance

Sep 2021 – Apr 2022

Prof. Satadal Ghosh, Aerospace Engineering, IIT Madras

Chennai, India

- Developed target-driven motion planning with real-time obstacle avoidance for autonomous unmanned aerial vehicles.
- Combined Proportional Navigation with Acceleration-Velocity Obstacles for reactive 2D guidance toward moving targets.
- Achieved roughly 20% faster and more compute-efficient trajectories than classical aerospace guidance algorithms.

Technical Projects

Multi-Agent Path Finding and Coordination | DD2438, KTH

Feb 2025 – May 2025

- Coordinated 20+ holonomic drones and non-holonomic cars with Conflict-Based Search over a Hybrid A* planner.
- Designed a greedy goal-assignment heuristic minimizing combined travel and assignment cost across the fleet.
- Added goal-padding and node-pruning heuristics with a fail-safe expansion bound for robust large-team planning.

Autonomous Mobile Manipulation Stack | DD2410, KTH

Sep 2024 – Oct 2024

- Built a Behavior Tree mission planner for a mobile manipulator handling navigation, grasping, and recovery in ROS.
- Implemented Hybrid A* and RRT planners for a Dubins car with collision checking against line and circle obstacles.
- Generated 2D occupancy grids from laser scans with raytraced free space and configuration-space inflation.

Hyperloop Pod Battery Pack and BMS | Avishkar Hyperloop, IIT Madras

Oct 2020 – Jun 2021

- Designed a high-discharge battery pack, extending pack life 10% by cutting peak temperature 10C with aluminium sinks.
- Built a custom Battery Management System for real-time voltage and temperature monitoring with safety cutoffs.
- Won the Most Scalable Design award and reached the top 24 teams for European Hyperloop Week in Valencia, Spain.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26**).(Under Review)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25 Workshop**)